



AI-Powered 3D Hairstyle Recommendation Platform

Innovation Project Document

Project Team

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01 PROBLEM STATEMENT

Millions of people visit salons every day hoping to get a trendy haircut they saw on social media — only to find that the hairstyle does not suit their face shape, head structure, or facial features. This mismatch leads to:

- **Dissatisfaction and regret** after spending money on an unflattering haircut.
- **Low self-confidence** caused by an appearance that does not meet expectations.
- **Financial waste**, as corrective haircuts or treatments are often needed afterward.
- **Communication gap** between customer and barber — customers cannot accurately visualize how a style will look before committing.

Currently, no accessible tool allows a person to preview how a specific hairstyle would look on **their own face and head shape** before visiting the salon. Existing solutions are limited to simple 2D overlays or generic filters that do not account for 3D head geometry, skin tone, or facial proportions.

02 PROPOSED SOLUTION

HAIRGURU is a web-based platform that uses Artificial Intelligence and 3D modeling to recommend the most suitable hairstyles for any user based on their facial photograph.

How It Works — User Flow

Step	Action	Description
1	Upload a Photo	User uploads a clear, front-facing photograph via the HAIRGURU website.
2	AI Face Analysis	The system detects face shape, proportions, and key facial landmarks.
3	3D Head Generation	HAIRGURU generates a realistic 3D model of the user's head.
4	Hairstyle Recommendation	The AI engine matches face shape to a curated hairstyle database.
5	3D Preview	Recommended styles are rendered; users can rotate and view from all angles.
6	Save & Share	Users save preferred looks or share them directly with their stylist.

03 INNOVATIVE IDEA

Comparison with Existing Solutions

Feature	Existing Apps	HAIRGURU
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Face shape detection	Basic or none	AI-powered precise classification
3D head modeling	Not available	Full 3D from a single photo
Recommendation logic	User manually browses	AI-driven face-shape matching
Multi-angle preview	Front-facing 2D overlay only	360° rotatable 3D preview
Personalization	Generic one-size-fits-all	Tailored to individual geometry

Key Innovations

- 1. **Single-Image 3D Reconstruction** — Generating a 3D head model from just one photograph, eliminating the need for expensive 3D scanners.
- 2. **AI-Driven Compatibility Matching** — The system intelligently suggests hairstyles proven to complement specific face shapes.
- 3. **Accessible to Everyone** — No app installation needed; works directly in the browser on any device.

04 SDG GOAL ALIGNMENT

SDG	Goal	How HAIRGURU Contributes
SDG 3	Good Health & Well-Being	Helps users choose flattering hairstyles, improving self-esteem and psychological well-being.
SDG 9	Industry, Innovation & Infrastructure	Applies cutting-edge AI & 3D modeling to innovate in the personal-care industry.
SDG 12	Responsible Consumption & Production	Reduces wasteful salon visits and corrective treatments through informed personal choices.

05 MINIMUM VIABLE PRODUCT (MVP)

#	Feature	Description
1	Photo Upload	User uploads a clear front-facing photo via the web interface.
2	Face Shape Detection	AI analyzes the photo to classify the user's face shape.
3	3D Head Model Generation	Generate a basic 3D head model from the uploaded photograph.
4	Hairstyle Recommendation	Display top 3–5 recommended hairstyles based on face shape.
5	3D Hairstyle Preview	Render recommended hairstyles on the 3D head model.
6	Responsive Web Interface	Clean, mobile-friendly website accessible from any device.

06 FUTURE ENHANCEMENTS

- 1. Hair Color Simulation — Preview different hair colors on the 3D model.
- 2. Beard & Facial Hair Recommendations — Extend to beard and mustache styles.
- 3. AR Live Preview — Real-time hairstyle overlay using the device camera.
- 4. Salon Integration — Book appointments directly with chosen style.
- 5. User Accounts & History — Save past recommendations and track preferences.
- 6. Social Sharing — Share 3D previews on social platforms.
- 7. Stylist Dashboard — Portal for professionals to view client preferences.
- 8. Multi-Language Support — Nepali, Hindi, and other regional languages.
- 9. Skin Tone & Texture Analysis — More holistic grooming recommendations.
- 10. Community & Reviews — User ratings and community-driven hairstyle database.

07 TARGET AUDIENCE

Audience Segment	Description
General Public	Anyone looking for a new hairstyle who wants to preview before committing.
Young Adults (18–35)	Social media-influenced individuals who frequently experiment with hairstyles.
Salon Customers	People visiting barbers/salons who want better communication with stylists.
Barbers & Stylists	Professionals who can use the tool to understand client expectations.
Fashion-Conscious Users	Individuals who value personal grooming and want data-driven style advice.

08 TECHNOLOGY STACK

Category	Technology / Tool
Frontend	React.js / Next.js, HTML5, CSS3, JavaScript
Backend	Python (Flask or FastAPI)
AI / ML	TensorFlow / PyTorch, OpenCV, Dlib, MediaPipe
3D Modeling	Three.js (web rendering), 3D Morphable Models (3DMM), PIFuHD
Face Analysis	CNN-based face shape classifier (custom trained)
Database	PostgreSQL / MongoDB
Cloud / Hosting	AWS / Google Cloud / Vercel
Version Control	Git & GitHub
Design/Prototyping	Figma

Project Management	Trello / Notion
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09 PROJECT TIMELINE & MILESTONES

Phase	Timeline	Milestone
Phase 1	Week 1–2	Research & requirement gathering; finalize tech stack
Phase 2	Week 3–4	Dataset collection (face images, hairstyle catalog)
Phase 3	Week 5–7	Train face shape classification model
Phase 4	Week 6–8	Develop 3D head reconstruction pipeline
Phase 5	Week 8–10	Build recommendation engine and hairstyle database
Phase 6	Week 9–11	Frontend development and API integration
Phase 7	Week 12	Testing, bug fixing, and optimization
Phase 8	Week 13–14	MVP demo, presentation, and documentation

10 POTENTIAL CHALLENGES & MITIGATION

Challenge	Mitigation Strategy
3D reconstruction accuracy from single photo	Use pre-trained models (3DMM, DECA) and fine-tune on diverse datasets
Face shape classification accuracy	Collect diverse training data across ethnicities; apply transfer learning.
Limited computational resources	Use cloud-based GPU instances (Google Colab, AWS) for model training.
Hairstyle database creation	Start with 50–100 curated styles; expand iteratively.
Real-time 3D rendering in browser	Optimize Three.js rendering; use Level of Detail (LOD) techniques.
Privacy and data security	Server-side processing with encryption; do not store photos permanently.
Team skill gaps in AI/3D	Leverage online courses, tutorials, and open-source pre-trained models.

11 MARKET RESEARCH

Industry Overview

The global hair care market was valued at over **\$87 billion in 2023** and is projected to grow at a CAGR of 5.5% through 2030. The virtual try-on market (AR/AI-based) is rapidly expanding, with companies like L'Oréal, ModiFace, and YouCam already investing heavily in AI beauty tools.

Competitor Analysis

Competitor	What They Offer	HAIRGURU's Advantage
YouCam Makeup	2D AR hairstyle try-on	Full 3D modeling, not just overlay
ModiFace (L'Oréal)	Color and makeup try-on	Focuses on hairstyle fit via face-shape AI
FaceApp	AI-based style filters	Personalized recommendations, not generic filters
Hair Zapp	Basic face shape quiz + suggestions	AI-precise face analysis, not manual quizzes

Gap in the Market

No existing platform combines **AI face shape analysis + 3D head reconstruction + personalized hairstyle recommendation** into a single, free, web-based tool. HAIRGURU fills this gap.

12 CONCLUSION

HAIRGURU addresses a real, everyday problem faced by millions — choosing the right hairstyle. By combining AI-powered face analysis with 3D head reconstruction, our platform empowers users to make confident, informed grooming decisions before ever stepping into a salon. The project is technically innovative, socially impactful (aligned with SDG 3, 9, and 12), and commercially viable in the growing virtual try-on market.

Document prepared by **Team HAIRGURU** — 2nd Semester Innovation Project
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